SEAA 4243 FINITE ELEMENT METHOD · 2L + 1T

2D Heat Transfer
Part 1 of 2

From the energy balance to the shape functions of the four-node scalar element

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Text: Mohd Yassin et al., *Two-Dimensional Finite Element Formulations for Heat, Solids and Fluids (with Matlab)*, MSNM 2020 — Chapter 2

COURSE MAP · AFTER THE MID-SEMESTER BREAK

Two topics in five sessions, then one session with ANSYS

1	2	3	4	5	6	7
Heat transfer 1 PDE · BCs · shape functions	**Heat transfer 2** [k] · assembly · solve · Asg 4	**Plane stress 1** PDE · [E] · [N]	**Plane stress 2** $[k] = \int [B]^T [E] [B]$ · loads	**Plane stress 3** stress recovery · Q8 · Project brief	**ANSYS demo** hand calc vs software	**Practical FEM** convergence · interpretation

CLO3 · the heat-transfer and plane-stress sessions

Formulate 2D finite element problems and solve by hand calculation and computer software.

CLO4 · project sessions

Investigate and discuss complex structural problems using FEM with the aid of computer software (group project).

Assessment on this block

Assignment 4 (heat transfer) released in Heat Transfer Part 2 · Project briefing Plane Stress Part 3 · Final exam covers CLO3.

LEARNING OUTCOMES · TODAY

By the end of this session you should be able to

1. Derive the steady 2D heat conduction PDE from an energy balance and Fourier's law, and write it as $\{\partial\}[E]\{\partial\}^T T = -q_h$.
2. Classify boundary conditions as Dirichlet (essential) or Neumann (natural) and write them for a rectangular domain.
3. Derive the bilinear shape functions $N_1 \ldots N_4$ of the four-node element and use them to interpolate $T(x, y)$.
4. Differentiate the approximation to obtain the temperature gradient and the heat flux at any point of an element.

WHY Outcomes 1–3 are the first half of Chapter 2 of the text (§2.1 – §2.2.1). Outcome 4 prepares the next session's conductance matrix [k].

SESSION MAP · 180 MINUTES

Six blocks, six short activities, one break

0:00	0:25	1:00	1:25	1:35	2:10	2:35
Why 2D?	**Governing PDE**	**Boundary conditions**	**Break**	**Shape functions**	**Gradient & flux**	**Pipeline & wrap-up**
25 min	35 min	25 min	10 min	35 min	25 min	25 min
A1 · 5 min	**A2 · 8 min**	**A3 · 6 min**	no activity	**A4 + QR**	**A5 · 8 min**	**Exit ticket**

How the activities work

- Amber slide = your turn. Work in pairs on paper, 5–8 minutes.
- A timer badge shows the time; the answer slide follows.
- Keep the answers — Activities 4 and 5 are the pattern for Assignment 4.

What to have ready

- Calculator, pen and the formula sheet (shape functions).
- The text, Chapter 2, open at Fig. 2.2.
- Next session: laptop with MATLAB for code tracing.

1

BLOCK 1 · 0:00 – 0:25

Why two dimensions?

FEM is a tool for solving a differential equation — so every topic starts from the PDE. Today’s PDE has one unknown: temperature.

BLOCK 1 · THE IDEA THAT ORGANISES THE COURSE

FEM is “nothing”: a tool that solves a differential equation

1 · Get the DE

Energy balance + Fourier’s law → a PDE in T(x, y)

2 · Discretise with FEM

Shape functions + Galerkin → $[k]\{\hat{T}\}^T = \{r\}$

3 · Solve

Assemble, impose $\bar{T}$, solve, post-process

FEM can be “everything”

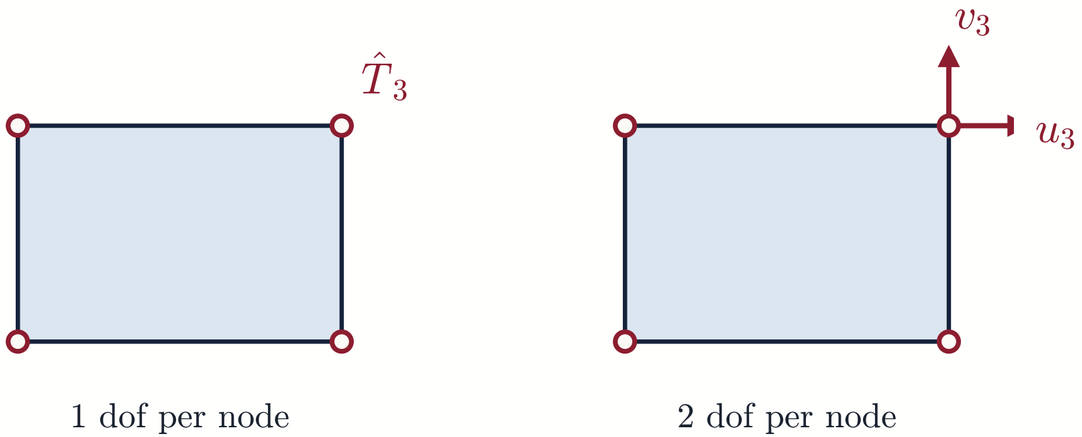
Heat, seepage, stress, vibration, fluid flow — the same machinery solves all of them. That is why it is worth learning well.

...and FEM can be “nothing”

It is only a way to turn a continuous PDE into simultaneous algebraic equations in matrix form. Nothing is new in the physics.

Finite difference, finite volume, meshless and isogeometric methods share the same goal; they differ in how the discretisation is done.

Heat transfer has one unknown per node; plane stress has two



Scalar element: each node carries one degree of freedom, the temperature $\hat{T}_i$. Vector element (Plane Stress Part 1): two, the displacements u_i and v_i.

Problems governed by the same scalar PDE

Field T	Property k	Flux q
temperature	conductivity	heat flux
hydraulic head	permeability	seepage velocity
electric potential	conductivity	current density
Prandtl stress function	1/G	shear stress

WHY Learn the scalar element once and you have also learnt seepage under a dam, groundwater flow and torsion of a shaft.

The same PDE appears across civil engineering

Mass concrete

Heat of hydration in dams, rafts and pile caps: the temperature rise and the gradient between core and surface drive early-age cracking.

Fire engineering

Temperature rise in steel sections and tunnel linings decides when strength is lost; the input is a 2D conduction model of the section.

Pavements and decks

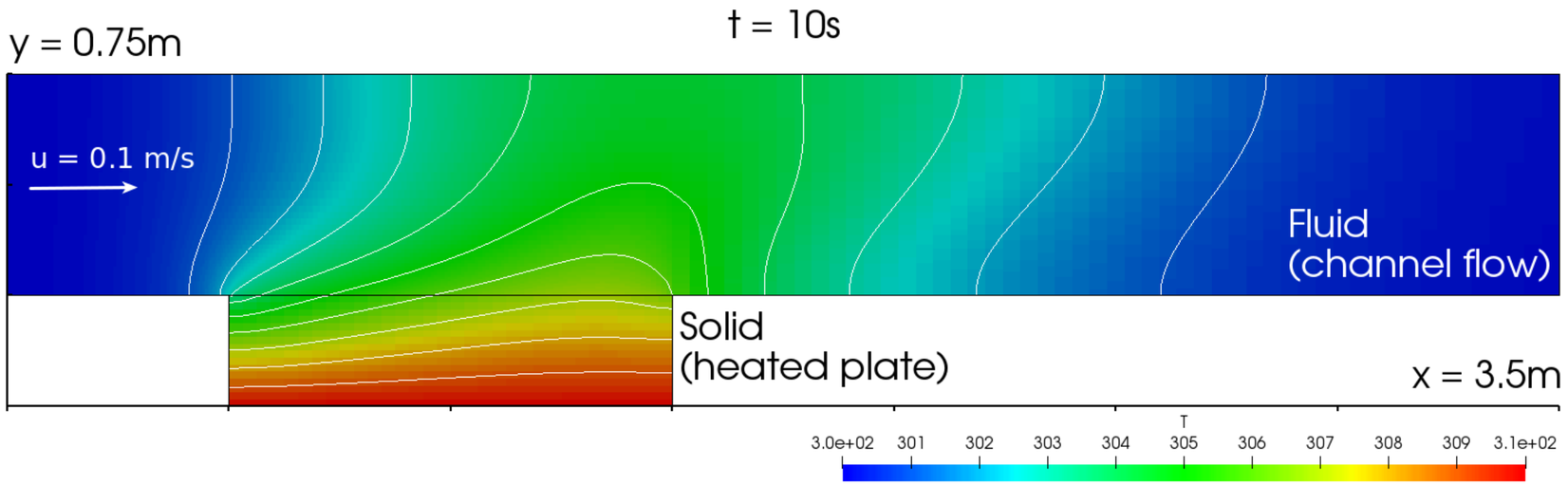
Daily thermal gradients through a bridge deck or a pavement slab produce stresses that codes make us check.

Ground and geothermal

Ground freezing, energy piles and buried pipelines are conduction problems in soil — and seepage uses the same equation.

BLOCK 1 · THE SAME PDE IN A REAL SIMULATION

A heated plate under a channel flow



WHY The solid plate obeys today's conduction PDE (with $\partial T/\partial t$ kept; at steady state it is exactly $\{\partial\}[E]\{\partial\}^T T = -q_h$), with T fixed on its heated bottom face.

CHECK At the interface the solid receives a temperature from the fluid code and returns a heat flux: one Dirichlet and one Neumann condition, as in Block 3.

Image: preCICE tutorials, "Flow over a heated plate", t = 10 s; solid by FEniCS, Nutils or CalculiX (FE). LGPL-3.0, github.com/precice/tutorials.

ACTIVITY 1 · PAIRS · NO CALCULATION

Scalar or vector? Classify each field and count the unknowns per node

5 MIN

Six fields — decide: scalar (1 dof) or vector (2 dof in 2D)?

- Temperature in a concrete slab
- Pore-water pressure in an embankment
- Displacement of a shear wall
- Velocity of water in a pipe network
- Electric potential in a grounding grid
- Settlement of a strip footing (2D section)

Then answer

- How many unknowns does a 4-node element have in each case?
- Which of the six could be solved with today’s element?

Compare with the pair next to you. We reveal in 5 minutes.

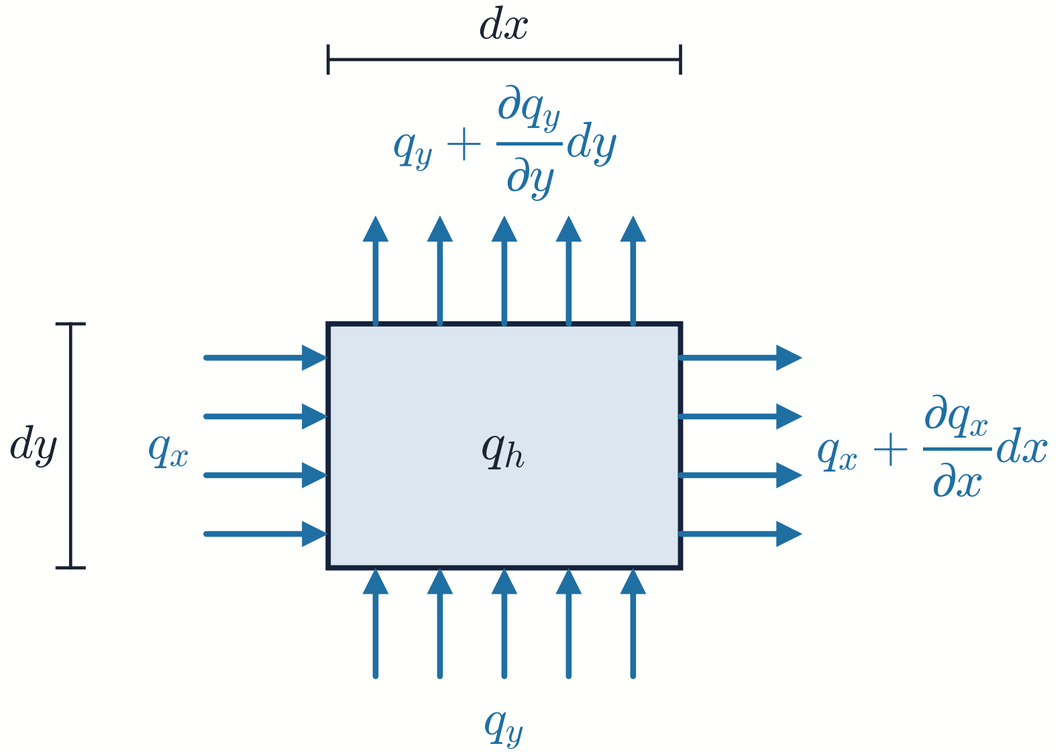
2

BLOCK 2 · 0:25 – 1:00

The governing PDE

Energy balance on a differential element, then Fourier’s law to close it in terms of temperature. Text §2.1.1 – §2.1.2.

Heat enters and leaves a small rectangle dx by dy



Dependent variables: the heat fluxes q_x and q_y (W m^{-2}), i.e. energy per unit time per unit area crossing a face.

The flux on the far face is the near-face flux plus its change across the element — a first-order Taylor expansion:

$$Q_x = q_x\,dy - \left(q_x + \frac{\partial q_x}{\partial x}dx\right)dy = -\frac{\partial q_x}{\partial x}\,dx\,dy \quad (2.2)$$

Inside, heat is generated at a rate q_h per unit volume (W m^{-3}), e.g. hydration of cement.

WHY Per unit thickness: t multiplies every term and cancels, so we take t = 1.

Rate of change of stored energy = net conduction in + heat generated

Stored (internal) energy

$$Q_E = \rho C \frac{\partial T}{\partial t} dx\, dy \quad (2.1)$$

ρ density, C specific heat, T temperature

Net conduction in

$Q_x + Q_y$, the x- and y-direction net inflows (2.2)–(2.4)

Generated

$$Q_h = q_h\, dx\, dy \quad (2.5)$$

$$\rho C \frac{\partial T}{\partial t} dx\, dy = -\frac{\partial q_x}{\partial x} dx\, dy - \frac{\partial q_y}{\partial y} dx\, dy + q_h\, dx\, dy \quad (2.6)$$

Every term carries the factor dx dy — the volume of the element per unit thickness — so it cancels (next slide).

ACTIVITY 2 · PAIRS · DERIVATION

Your turn: the y-direction balance, then cancel dx dy

8 MIN

Do

- Write Q_y by the same argument as Q_x (in through y = 0, out through y = dy).
- Put $Q_E = Q_x + Q_y + Q_h$ and cancel the common factor dx dy.
- Check the units of every term in the result.

$$Q_x = q_x\,dy - \left(q_x + \frac{\partial q_x}{\partial x}dx\right)dy = -\frac{\partial q_x}{\partial x}\,dx\,dy \quad \text{(2.2) — given}$$

$$Q_E = \rho\,C\,\frac{\partial T}{\partial t}\,dx\,dy \quad \text{(2.1) — given}$$

CHECK Target: an equation with ∂T/∂t on the left and three terms on the right, none containing dx or dy.

Reveal follows. If you finish early: what changes if q_h depends on T?

ACTIVITY 2 · SOLUTION

The energy balance with dx dy cancelled

$$Q_y = q_y\,dx - \left(q_y + \frac{\partial q_y}{\partial y}dy\right)dx = -\frac{\partial q_y}{\partial y}\,dx\,dy \quad (2.3)$$

$$\rho C\frac{\partial T}{\partial t}\,dx\,dy = -\frac{\partial q_x}{\partial x}\,dx\,dy - \frac{\partial q_y}{\partial y}\,dx\,dy + q_h\,dx\,dy \quad (2.6)$$

$$\rho C\frac{\partial T}{\partial t} = -\frac{\partial q_x}{\partial x} - \frac{\partial q_y}{\partial y} + q_h \quad (2.7)$$

CHECK Units of every term: W m^{-3}. If one term disagrees, the derivation has a mistake.

WHY (2.7) still has three unknowns q_x, q_y, T. We need a law relating flux to temperature.

BLOCK 2 · FOURIER'S LAW CLOSES THE SYSTEM

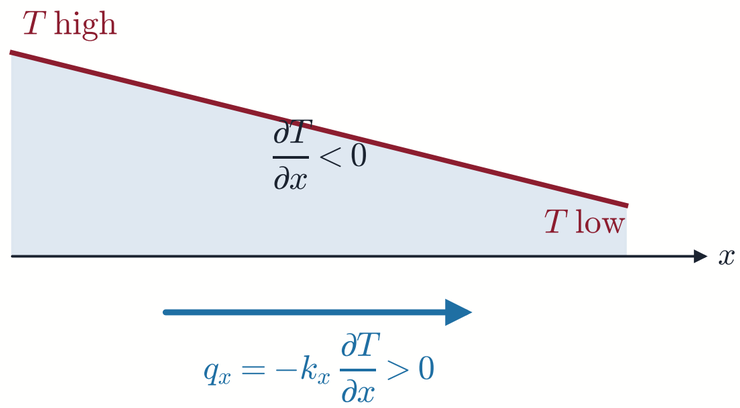
Heat flows down the temperature gradient

$$q_x = -k_x \frac{\partial T}{\partial x}, \qquad q_y = -k_y \frac{\partial T}{\partial y} \quad (2.8), (2.9)$$

k_x, k_y: thermal conductivity in W m^{-1} K^{-1}. Isotropic material: $k_x = k_y = k$ (concrete ≈ 1.5–2.5, steel ≈ 45–50).

Substituting (2.8)–(2.9) into (2.7) gives the general (unsteady) conduction equation:

$$\rho C \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_y \frac{\partial T}{\partial y}\right) + q_h \quad (2.10)$$



WHY The minus sign: where T falls with x ($\partial T/\partial x < 0$) the flux is positive, i.e. towards +x. Heat goes from hot to cold.

Drop the time derivative: the PDE we discretise

$$\frac{\partial}{\partial x}\left(k_x \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k_y \frac{\partial T}{\partial y}\right) = -q_h \quad (2.11)$$

Special cases you already know

$$k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right) = -q_h \qquad \text{Poisson: constant isotropic } k$$

$$\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} = 0 \qquad \text{Laplace: no source}$$

WHY q_h sits on the right with a minus sign because we moved it across; a source ($q_h > 0$) makes the field concave — a hill.

CHECK Unit check of each term of (2.11): W m^{-3}. The equation says: divergence of the flux = source.

Write the operator as a row vector times [E] times a column vector

$$\left\{ \frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \right\} \begin{bmatrix} k_x & 0 \\ 0 & k_y \end{bmatrix} \left\{ \begin{matrix} \dfrac{\partial}{\partial x} \\ \\ \dfrac{\partial}{\partial y} \end{matrix} \right\} T = -q_h \qquad (2.12)$$

$$\{\partial\}\,[E]\,\{\partial\}^T\,T = -q_h \qquad (2.13)$$

$$\{\partial\} = \left\{ \frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \right\}, \qquad [E] = \begin{bmatrix} k_x & 0 \\ 0 & k_y \end{bmatrix} \qquad (2.14)$$

WHY Why bother? In Plane Stress Part 1 plane stress will read [∂][E][∂]ᵀ{u}ᵀ = –{f}: the same shape with a 3 × 3 [E] and a 2 × 3 operator. Learn the pattern once.

WATCH OUT {∂} is a row of operators, not numbers: it acts on whatever stands to its right. Order of multiplication matters.

BLOCK 3 · 1:00 – 1:25

3

Boundary conditions

A PDE alone is not a problem. Specify either the temperature or the flux on every part of the boundary. Text §2.1.3.

Either T or the normal flux is known at every boundary point

Dirichlet · essential · temperature specified

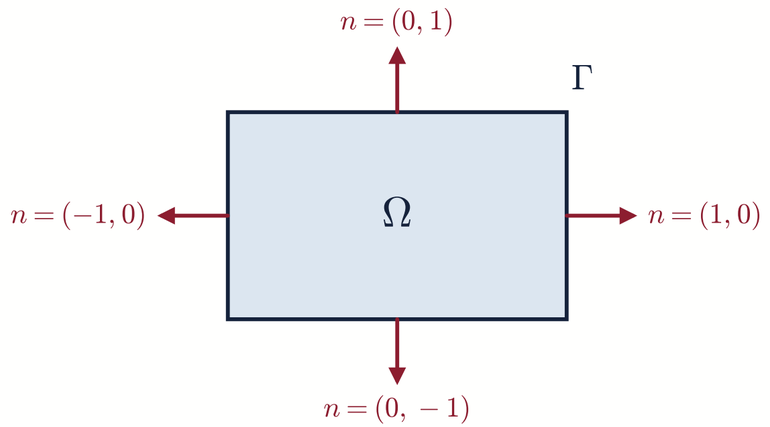
$$T\big|_b = \bar{T} \quad (2.16)$$

e.g. a face held at a fixed temperature by a fluid bath or the ground.

Neumann · natural · normal flux specified

$$\left(k_x \frac{\partial T}{\partial x}\Big|_b\right) n_x + \left(k_y \frac{\partial T}{\partial y}\Big|_b\right) n_y = q_b \quad (2.15)$$

q_b known flux; (n_x, n_y) outward unit normal. Convection and radiation are Neumann-type too (not treated here).



WATCH OUT Well-posed: the whole boundary is covered, and at least one point has T specified — otherwise T is only known up to a constant.

BLOCK 3 · RECTANGULAR DOMAINS

On a rectangle the normal is a coordinate axis, so (2.15) collapses to one term

$$n = (\pm 1,\, 0): \quad \pm k_x \left.\frac{\partial T}{\partial x}\right|_b = q_b$$ vertical edges

$$n = (0,\, \pm 1): \quad \pm k_y \left.\frac{\partial T}{\partial y}\right|_b = q_b$$ horizontal edges

$$\text{insulated edge:} \quad q_b = 0 \;\Rightarrow\; \left.\frac{\partial T}{\partial n}\right|_b = 0$$ insulated

WATCH OUT Sign: $q_b = k\,\partial T/\partial n$ is positive when heat **enters**. On the right edge n = (+1, 0): an inward flux makes $\partial T/\partial x > 0$ there, so $q_b > 0$ — while the flux vector component $q_x = -k\,\partial T/\partial x$ is negative. Two different symbols; keep them apart.

WHY Insulated = adiabatic = zero normal flux. It is a Neumann condition with $q_b = 0$, not a Dirichlet condition.

BLOCK 3 · WHAT FEM DOES WITH EACH KIND

Essential conditions are imposed; natural conditions appear by themselves

Condition	What is known	Where it enters the FEM	Next session
Dirichlet · essential	$\bar{T}$ on the edge	imposed on the assembled system by modifying rows and columns of [K]	row–column elimination
Neumann · natural	q_b on the edge	appears in the boundary vector {b} produced by integration by parts	$\{b\} = \int \{N\}^T q_n \, ds$
Insulated · natural, $q_b = 0$	no flux	contributes nothing: the boundary integral is zero	do nothing

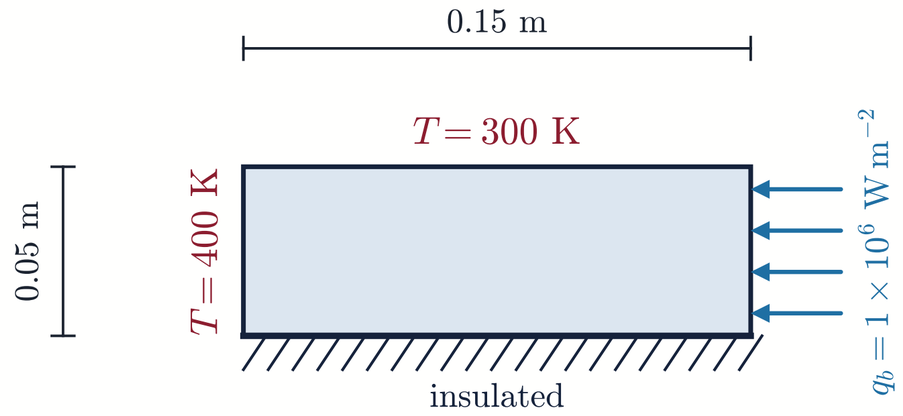
WATCH OUT The most common error in Assignment 4: trying to “impose” an insulated edge. It needs no action at all.

WHY The names come from the integration by parts: the natural condition is the term that falls out of it; the essential one must be enforced on the unknowns.

ACTIVITY 3 · PAIRS · CLASSIFICATION

Label the plate: for each edge, type, equation and what FEM does

6 MIN



For each of the four edges write

- Dirichlet or Neumann?
- The equation, with the outward normal and the correct sign.
- Imposed, becomes a load, or do nothing?

Then: is the problem well-posed? Why?

Text Exercise 2.5 geometry: k = 450 W m^{-1} K^{-1}, q_h = 1 × 10^7 W m^{-3}, unit thickness.

Reveal follows — then a 10-minute break.

ACTIVITY 3 · SOLUTION

Two essential edges, one flux load, one edge that needs nothing

Edge	Type	Equation	FEM action
top	Dirichlet	$T = 300$ K	impose on the system
left	Dirichlet	$T = 400$ K	impose on the system
right	Neumann	$n = (1, 0)$: $k\,\partial T/\partial x = q_b = +1 \times 10^6$ W m^{-2} (inward, so $q_x = -10^6$)	load in {b}
bottom	Neumann, $q_b = 0$	$\partial T/\partial y = 0$	nothing

CHECK Well-posed: every edge has a condition and T is fixed on two edges, so the temperature level is pinned.

WHY q_b is the boundary flux of (2.15), positive inward; the flux-vector component q_x points along $-n$ and is negative.

1:25 – 1:35

Break

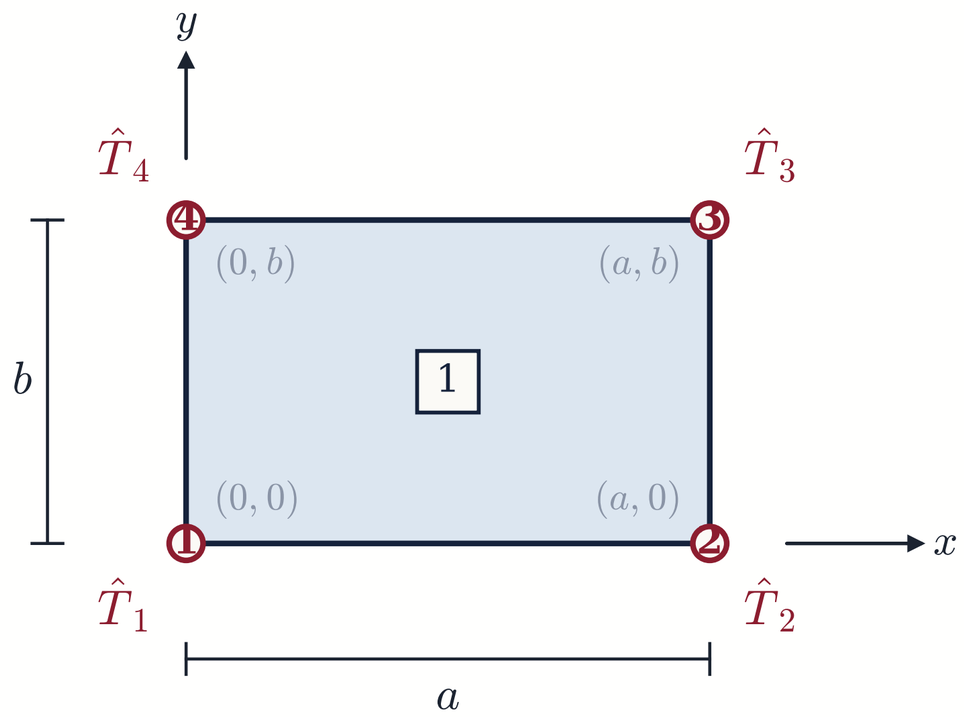
Back in 10 minutes. Second half: shape functions, gradients, and the pipeline for the next session.

BLOCK 4 · 1:35 – 2:10

4 The four-node scalar element

One unknown per node, a bilinear polynomial in between. Derive $N_1 \ldots N_4$ and learn the three properties you can always check. Text §2.2.1.

A rectangle a × b, nodes numbered anticlockwise



Notation

- $\hat{T}_i$ (hat): the nodal temperature at node i — a number, the unknown we solve for.
- T(x, y): the temperature field inside the element — a function.
- $N_i(x, y)$: the shape function that carries the influence of node i into the field.
- Local coordinates: origin at node 1, x along the bottom edge, y along the left edge.

WATCH OUT Numbering is anticlockwise. Reverse it and the matrix entries the next session come out in the wrong places.

BLOCK 4 · ASSUME A POLYNOMIAL · TEXT (2.17)

Four nodes, four coefficients: 1, x, y and xy

$$T = a_1 + a_2\,x + a_3\,y + a_4\,xy \quad (2.17)$$

Why these four monomials?

- Four nodal values can fix exactly four coefficients.
- 1, x, y give a plane; xy is the lowest-order term that is symmetric in x and y.
- x^2 without y^2 would favour one direction — the choice is geometrically balanced.
- Along any edge (x = const or y = const) the polynomial is linear in the other variable.

Pascal triangle · text Fig. 2.4

$$1$$
$$x \quad y$$
$$x^2 \quad xy \quad y^2$$
$$x^3 \quad x^2y \quad xy^2 \quad y^3$$

4-node element: the four maroon terms. 8-node element: all terms down to x^2y and xy^2 (Block 5).

WHY “Bilinear”: linear in x for fixed y and linear in y for fixed x — but not a plane, because of the xy term.

BLOCK 4 · EVALUATE AT THE NODES · TEXT (2.18)

Each nodal value gives one equation in the coefficients

$$
\begin{aligned}
T\big|_{x=0,\,y=0} &= a_1 = \hat{T}_1 \\
T\big|_{x=a,\,y=0} &= a_1 + a_2\,a = \hat{T}_2 \\
T\big|_{x=a,\,y=b} &= a_1 + a_2\,a + a_3\,b + a_4\,ab = \hat{T}_3 \\
T\big|_{x=0,\,y=b} &= a_1 + a_3\,b = \hat{T}_4
\end{aligned}
\tag{2.18}
$$

$$
\begin{bmatrix} 1 & 0 & 0 & 0 \\ 1 & a & 0 & 0 \\ 1 & a & b & ab \\ 1 & 0 & b & 0 \end{bmatrix}
\begin{Bmatrix} a_1 \\ a_2 \\ a_3 \\ a_4 \end{Bmatrix}
=
\begin{Bmatrix} \hat{T}_1 \\ \hat{T}_2 \\ \hat{T}_3 \\ \hat{T}_4 \end{Bmatrix}
$$

matrix form

WHY Lower-triangular after reordering, so it solves by forward substitution: a_1 first, then a_2, a_3, finally a_4.

Four coefficients in terms of the four nodal temperatures

$$a_1 = \hat{T}_1, \qquad a_2 = \frac{-\hat{T}_1 + \hat{T}_2}{a}, \qquad a_3 = \frac{-\hat{T}_1 + \hat{T}_4}{b}, \qquad a_4 = \frac{\hat{T}_1 - \hat{T}_2 + \hat{T}_3 - \hat{T}_4}{a\,b} \tag{2.19}$$

Read them physically

- a_1: the value at the origin (node 1).
- a_2: the slope along the bottom edge, $(\hat{T}_2 - \hat{T}_1)/a$.
- a_3: the slope along the left edge, $(\hat{T}_4 - \hat{T}_1)/b$.
- a_4: the “twist” — how the x-slope changes with y.

Check with a uniform field

Put $\hat{T}_1 = \hat{T}_2 = \hat{T}_3 = \hat{T}_4 = T_0$: then $a_1 = T_0$ and $a_2 = a_3 = a_4 = 0$, so $T = T_0$ everywhere. The element reproduces a constant exactly.

BLOCK 4 · COLLECT BY NODAL VALUE · TEXT (2.20)–(2.22)

The multiplier of each $\hat{T}_i$ is its shape function N_i

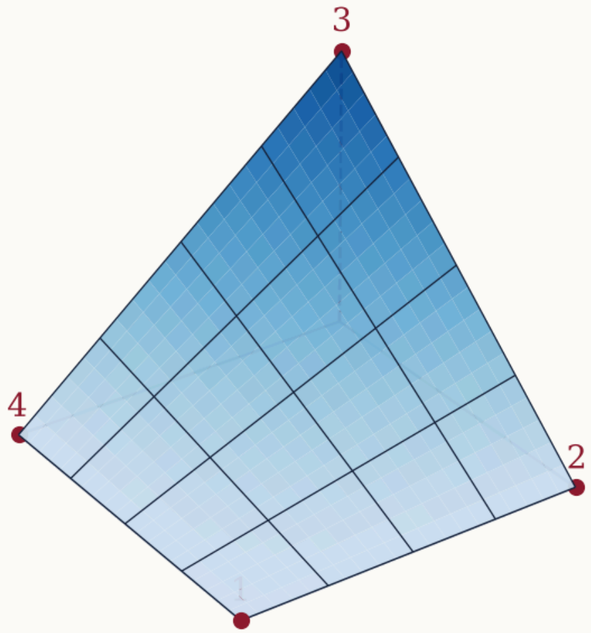
$$T(x,y) = \left(\frac{(a-x)(b-y)}{ab}\right)\hat{T}_1 + \left(\frac{x\,(b-y)}{ab}\right)\hat{T}_2 + \left(\frac{x\,y}{ab}\right)\hat{T}_3 + \left(\frac{y\,(a-x)}{ab}\right)\hat{T}_4 \quad (2.20)$$

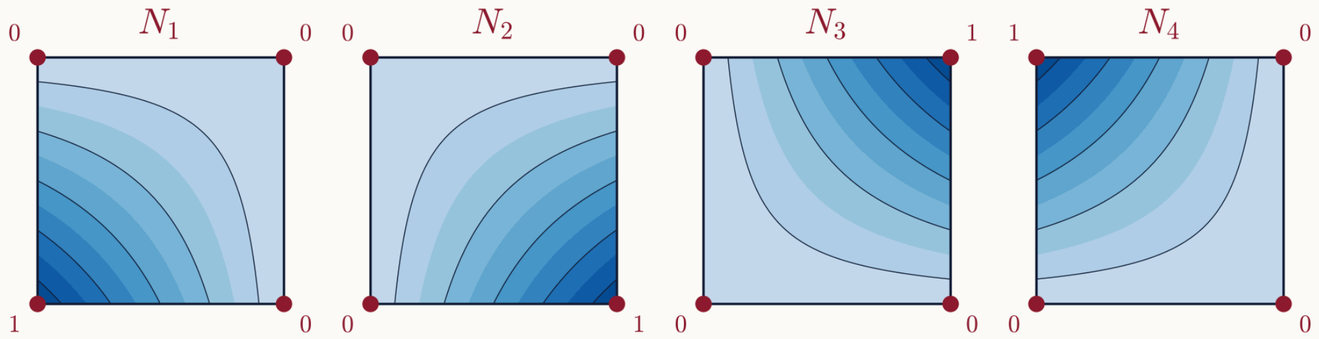
$$N_1 = \frac{(a-x)(b-y)}{ab}, \qquad N_2 = \frac{x\,(b-y)}{ab}, \qquad N_3 = \frac{x\,y}{ab}, \qquad N_4 = \frac{y\,(a-x)}{ab} \quad (2.21)$$

$$T = N_1\hat{T}_1 + N_2\hat{T}_2 + N_3\hat{T}_3 + N_4\hat{T}_4 = \{N\}\{\hat{T}\}^T \quad (2.22)$$

WHY {N} is a 1 × 4 row; $\{\hat{T}\}^T$ is a 4 × 1 column. Their product is the scalar T(x, y).

A tent: 1 at its own node, 0 at the other three, straight along every edge



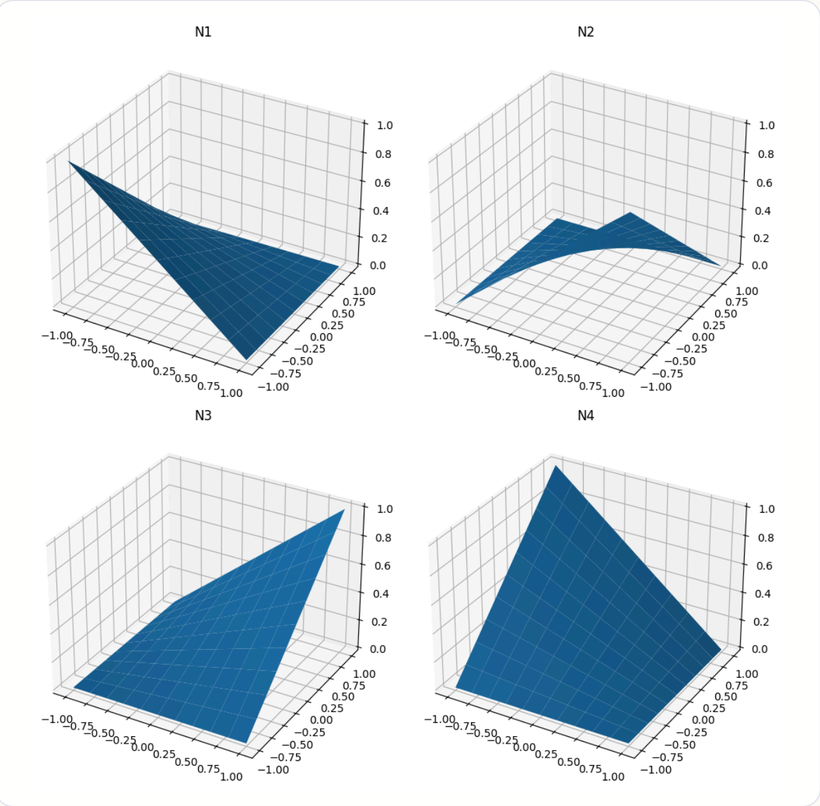


Left: N_3 as a surface. Right: all four as colour maps (dark = 1). Each N_i is the temperature field you get when node i is at 1 and the others at 0.

WHY Along an edge only the two nodes on that edge matter — the straight line of the bar element. That is what lets two neighbouring elements agree along their common edge.

BLOCK 4 · THE SAME FUNCTIONS INSIDE COMMERCIAL CODE

ANSYS's four-node element uses these N_1 … N_4



Same functions, new coordinates

ξ = 2x/a − 1 and η = 2y/b − 1 map 0 ≤ x ≤ a, 0 ≤ y ≤ b onto −1 ≤ ξ, η ≤ 1, and

$N_1 = \frac{1}{4}(1 - \xi)(1 - \eta) = (a - x)(b - y)/ab$

which is the text's (2.21). The other three follow by changing signs.

CHECK Each surface is 1 at its own corner, 0 at the other three and straight along every edge: the three checks on the next slide.

WHY Plane Stress Part 2: the square lets one formula serve any four-sided element — the isoparametric idea behind PLANE182.

Image: PyMAPDL documentation, "Build a finite element model" (how a finite element works), © ANSYS, Inc., MIT licence, mapdl.docs.pyansys.com.

BLOCK 4 · THREE PROPERTIES YOU CAN CHECK IN 30 SECONDS

Kronecker delta, partition of unity, linear edges

1 · Kronecker delta

$$N_i(x_j, y_j) = \delta_{ij} = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$$

So T at node j equals $\hat{T}_j$: the nodal unknowns really are nodal temperatures.

2 · Partition of unity

$$\sum_{i=1}^{4} N_i = 1 \quad \text{everywhere in the element}$$

A uniform temperature is reproduced exactly — the test we did on the coefficients.

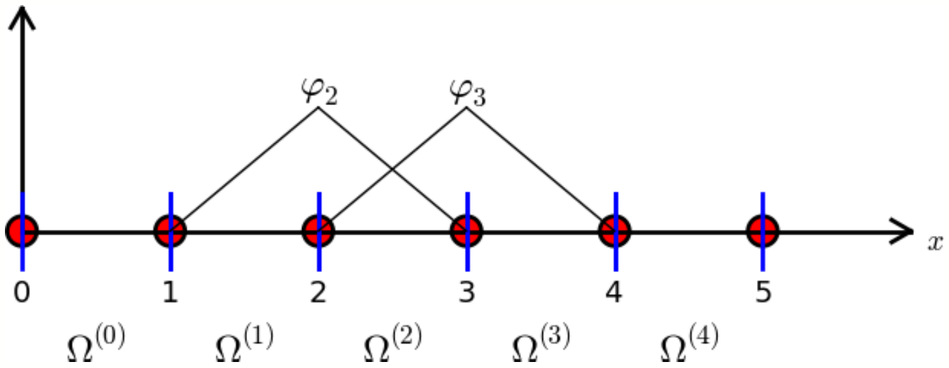
3 · Linear along each edge

$$\text{on } y = 0: \quad T = \left(1 - \frac{x}{a}\right)\hat{T}_1 + \frac{x}{a}\,\hat{T}_2 \quad \text{(linear, depends on 2 nodes only)}$$

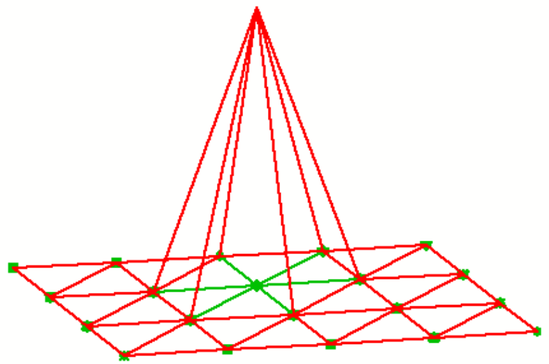
C^0 continuity follows: the neighbouring element shares this edge's two nodes and its straight line.

BLOCK 4 · FROM ONE ELEMENT TO THE MESH

One nodal value lifts one tent



1D: φ_i is a hat over the two elements that share node i.



2D: a tent over every element sharing the node.

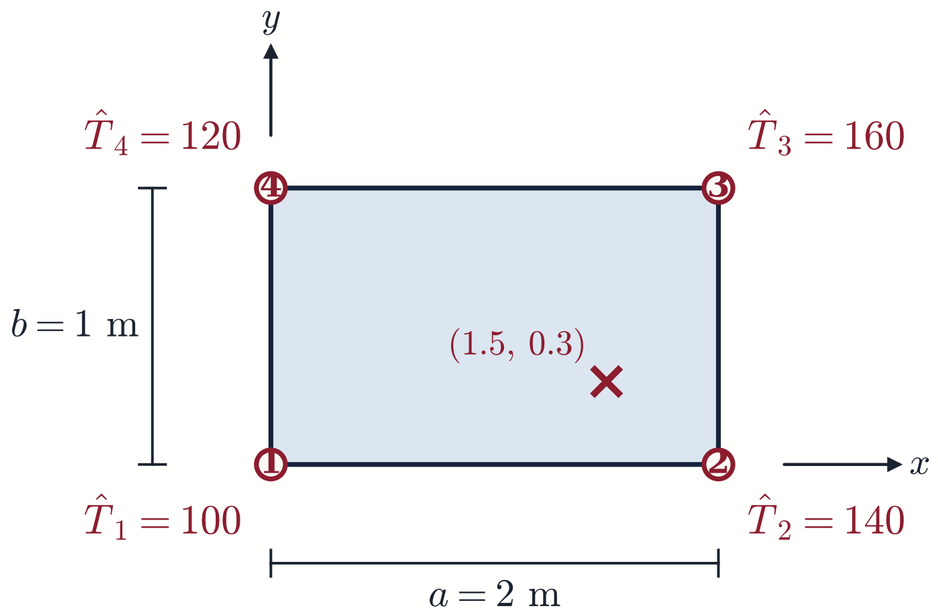
WHY Each element that touches node i adds one face of its tent. That is why T is continuous across element edges, and why node i couples only to its neighbours: the global matrix is sparse (Heat Transfer Part 2).

Images: Langtangen and Mardal, *Introduction to Numerical Methods for Variational Problems* (Springer, 2019), github.com/hplgit/fem-book, MIT licence.

ACTIVITY 4 · PAIRS · CALCULATION

Evaluate the shape functions at a point and interpolate the temperature

8 MIN



Given

- a = 2 m, b = 1 m; point (x, y) = (1.5, 0.3) m.
- Nodal temperatures $\hat{T}$ = {100, 140, 160, 120} K.

Find

- N_1, N_2, N_3, N_4 at the point — and their sum.
- T(1.5, 0.3).

CHECK Keep the sheet: Activity 5 uses the same element and data.

8 minutes. Early finishers: which node has the most influence at this point, and why?

ACTIVITY 4 · SOLUTION

Node 2 carries half the weight; the four weights sum to one

$$N_1 = \frac{(2-1.5)(1-0.3)}{2\cdot 1} = \frac{(0.5)(0.7)}{2} = 0.175, \qquad N_2 = \frac{x\,(b-y)}{ab} = \frac{(1.5)(0.7)}{2} = 0.525$$

$$N_3 = \frac{x\,y}{ab} = \frac{(1.5)(0.3)}{2} = 0.225, \qquad N_4 = \frac{y\,(a-x)}{ab} = \frac{(0.3)(0.5)}{2} = 0.075$$

(2.21) at (1.5, 0.3)

$$\sum N_i = 0.175 + 0.525 + 0.225 + 0.075 = 1.000$$ partition of unity

$$\begin{aligned} T &= 0.175(100) + 0.525(140) + 0.225(160) + 0.075(120) \\ &= 17.5 + 73.5 + 36.0 + 9.0 = 136.0\ \text{K} \end{aligned}$$

(2.22)

Assignment-type question: “after solving, $\hat{T}_2$ and $\hat{T}_6$ are 150 K … interpolate T at (1.4, 2.2)” — the same method, after converting the global point to local element coordinates.

QR EXPLORER · PAIRS, ONE PHONE

Check Activity 4 on your phone, then explore

6 MIN

Scan, then do

1. Press “Activity 4 point”. Do N_1 … N_4, their sum and T = 136.0 K match your sheet?
2. Drag P onto node 3 and read the four weights. Which property is this?
3. Find three points where $N_1 = N_3$. What line joins them?

Shape Function Explorer

`razinutm.github.io/My_Lectures/SEAA4243-FEM/labs/shape.html`

6 minutes, optional in class: if the block is running late, set it as homework. Answers are inside the Explorer under “Check answer”.

BLOCK 5 · 2:10 – 2:35

5

Gradient, flux and a look ahead

Differentiate the approximation: the 2 × 4 matrix $\{\partial\}^T\{N\}$ gives the gradient, Fourier gives the flux — and the next session it builds [k].

BLOCK 5 · DIFFERENTIATE THE APPROXIMATION

Gradient = derivative matrix × nodal temperatures

$$\begin{Bmatrix} \dfrac{\partial T}{\partial x} \\ \dfrac{\partial T}{\partial y} \end{Bmatrix} = \{\partial\}^T \{N\} \{\hat{T}\}^T = \begin{bmatrix} \dfrac{\partial N_1}{\partial x} & \dfrac{\partial N_2}{\partial x} & \dfrac{\partial N_3}{\partial x} & \dfrac{\partial N_4}{\partial x} \\ \dfrac{\partial N_1}{\partial y} & \dfrac{\partial N_2}{\partial y} & \dfrac{\partial N_3}{\partial y} & \dfrac{\partial N_4}{\partial y} \end{bmatrix} \begin{Bmatrix} \hat{T}_1 \\ \hat{T}_2 \\ \hat{T}_3 \\ \hat{T}_4 \end{Bmatrix}$$

$$\{\partial\}^T \{N\} = \frac{1}{ab} \begin{bmatrix} -(b-y) & (b-y) & y & -y \\ -(a-x) & -x & x & (a-x) \end{bmatrix}$$ from (2.21)

WHY $\partial N_i/\partial x$ depends on y only and $\partial N_i/\partial y$ on x only — the gradient varies inside the element, though each component is constant along one direction.

CHECK Each row of $\{\partial\}^T\{N\}$ sums to zero. A uniform $\hat{T}$ therefore gives zero gradient, as it must.

BLOCK 5 · FROM GRADIENT TO FLUX

Fourier’s law in matrix form is the post-processing formula

$$\begin{Bmatrix} q_x \\ q_y \end{Bmatrix} = -[E]\,\{\partial\}^T\{N\}\{\hat{T}\}^T$$

Once the nodal temperatures are known, this gives the heat flux at any point (x, y) of the element — the quantity a designer usually wants (heat loss through a wall, flow under a dam).

$$[k] = \int_x \int_y \{N\}^T\{\partial\}[E]\{\partial\}^T\{N\}\,dy\,dx$$

$$[k]\{\hat{T}\}^T = \{q\} + \{b\} = \{r\}$$

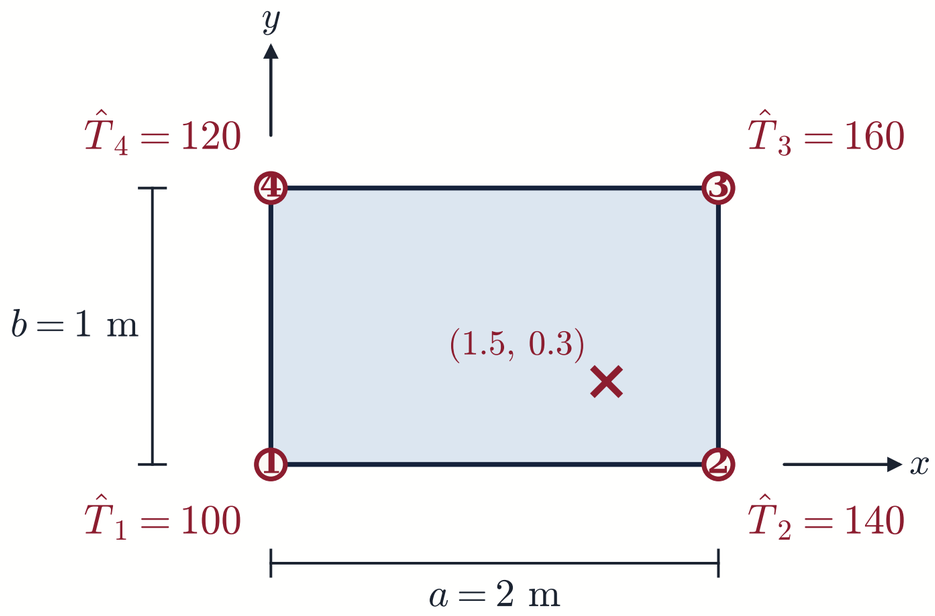
Next session: text (2.38) and (2.42).

WHY The same {∂}ᵀ{N} appears twice in [k], once transposed: [k] is symmetric, and its row sums are zero for the same reason the gradient of a uniform field is zero.

ACTIVITY 5 · PAIRS · CALCULATION

Heat flux at the same point: evaluate the derivatives, then apply Fourier

8 MIN



Given (as Activity 4)

- a = 2 m, b = 1 m, point (1.5, 0.3) m, $\hat{T}$ = {100, 140, 160, 120} K.
- $k_x = k_y = 2$ W m^{-1} K^{-1}.

Find

- $\partial N_i/\partial x$ for i = 1…4, then $\partial T/\partial x$.
- q_x from Fourier's law — sign and direction?
- If time: $\partial T/\partial y$ and q_y.

CHECK Check: the four derivatives must sum to zero.

8 minutes. Reveal follows.

ACTIVITY 5 · SOLUTION

Heat flows from the hot side (nodes 2, 3) towards the cold side

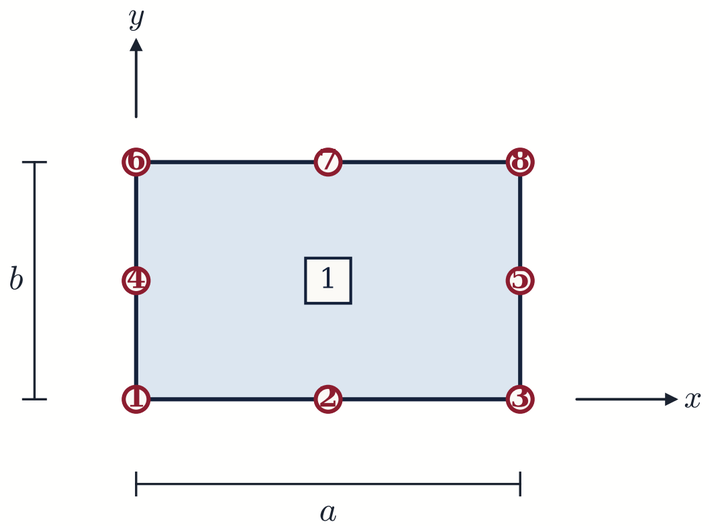
$$\frac{\partial N_i}{\partial x} = \frac{1}{ab}\{-(b-y),\ (b-y),\ y,\ -y\} = \{-0.35,\ 0.35,\ 0.15,\ -0.15\}$$

$$\begin{aligned}\frac{\partial T}{\partial x} &= \sum_i \frac{\partial N_i}{\partial x}\hat{T}_i = -0.35(100) + 0.35(140) + 0.15(160) - 0.15(120) \\ &= -35 + 49 + 24 - 18 = 20\ \mathrm{K\,m^{-1}}\end{aligned}$$

$$q_x = -k_x \frac{\partial T}{\partial x} = -(2)(20) = -40\ \mathrm{W\,m^{-2}}$$

$$\frac{\partial T}{\partial y} = -0.25(100) - 0.75(140) + 0.75(160) + 0.25(120) = 20\ \mathrm{K\,m^{-1}} \ \Rightarrow\ q_y = -40\ \mathrm{W\,m^{-2}}$$

Mid-side nodes, eight monomials, quadratic along each edge



$$T = a_1 + a_2 x + a_3 y + a_4 x^2 + a_5 y^2 + a_6 xy + a_7 x^2 y + a_8 x y^2 \quad (2.23)$$

$$\{N\} = \{x\}\,[\,x|\,]^{-1}, \qquad \text{e.g. } N_5 = \frac{4\,x\,(a-x)(b-y)}{a^2\,b} \quad (2.30),\ (2.31e)$$

- Same recipe: evaluate at 8 nodes, invert the 8 × 8 matrix, collect — the text gives $N_1 \ldots N_8$ in (2.31).
- Three nodes per edge, so T is quadratic along an edge: fewer elements for curved fields.
- Hand calculations here use the 4-node element; the 8-node one lives in the MATLAB code and in ANSYS (PLANE183-type).

BLOCK 6 · 2:35 – 3:00

6

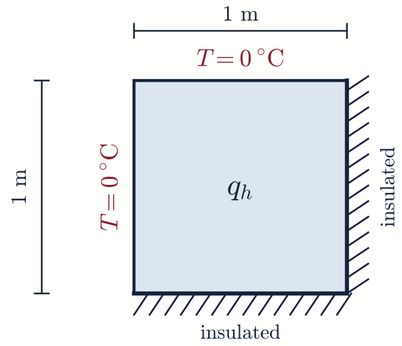
The pipeline — where today fits

Seven steps from PDE to answer. Today covered the first two; the next session we do the other five and run the MATLAB code.

BLOCK 6 · SEVEN STEPS FROM PDE TO ANSWER

Today: steps 1 and 2. Next session: steps 3 to 7 on the text's worked example

1	2	3	4	5	6	7
PDE + boundary conditions	**Shape functions {N}**	**Galerkin weighted residual**	**Integration by parts**	**Element matrices [k], {q}, {b}**	**Assembly + impose $\bar{T}$**	**Solve, then post-process**
today	today	Heat Transfer Part 2	Heat Transfer Part 2	Heat Transfer Part 2	Heat Transfer Part 2	Heat Transfer Part 2



The next session's worked example · text §2.3

Square plate 1 m × 1 m, q_h = 600 W m^{-3}, $k_x = k_y$ = 300 W m^{-1} K^{-1}; top and left edges at 0 °C, right and bottom edges insulated. Four 4-node elements by hand, then 64 elements in MATLAB, then software.

Before the next session, classify its four edges as in Activity 3 and predict where the hottest point is.

BLOCK 6 · THE MATLAB SKELETON FOLLOWS THE PIPELINE

Thirty lines: local matrix, assembly, boundary conditions, solve

```
% Local conductance matrix (text §2.3.1)
a = L/Nx; b = H/Ny;
s1 = b*kx/(3*a) + a*ky/(3*b);
s2 = -b*kx/(3*a) + a*ky/(6*b);
s3 = -b*kx/(6*a) - a*ky/(6*b);
s4 = b*kx/(6*a) - a*ky/(3*b);
k = [s1 s2 s3 s4; s2 s1 s4 s3; s3 s4 s1 s2; s4 s3 s2 s1];
fQ = a*b*Q/4*[1; 1; 1; 1]; % heat generation
% Assembly by connectivity, element 1 = nodes [1 2 5 4]
K([1 2 5 4],[1 2 5 4]) = K([1 2 5 4],[1 2 5 4]) + k;
F([1 2 5 4]) = F([1 2 5 4]) + fQ;
% Impose T = To on known dofs, solve the rest
dof_k = [1 4 7 8 9]; T(dof_k) = To;
F = F - K(:,dof_k)*T(dof_k);
dof_u = setdiff(1:ndof, dof_k);
T(dof_u) = K(dof_u,dof_u) \ F(dof_u);
```

Read the code against the steps

- Lines 2–8: step 5, the element matrix and load from closed-form integrals (we derive $s_1 \ldots s_4$ the next session).
- Lines 11–12: step 6, assembly — the connectivity list [1 2 5 4] is the element's global node numbers.
- Lines 15–18: step 6, essential conditions by row–column elimination; step 7, solve with the backslash operator.

CHECK Bring a laptop with MATLAB the next session; we trace this code line by line on the 2 × 2 mesh.

ACTIVITY 6 · INDIVIDUAL · EXIT TICKET

Three questions before you leave — on paper, hand in at the door

6 MIN

Questions

- Which boundary condition needs no action at all in the FEM, and why?
- For a square element (a = b), what is each N_i at the centroid? What is their sum?
- Write $\{\partial\}[E]\{\partial\}^T T = -q_h$ out in full for $k_x = k_y = k$. Which classical equation is it?

$$N_i\left(\tfrac{a}{2}, \tfrac{b}{2}\right) = \frac{(a/2)(b/2)}{ab} = \frac{1}{4} \quad \text{for every } i$$ hint for Q2

$$\{\partial\}[E]\{\partial\}^T T = -q_h \quad \text{with } k_x = k_y = k$$ Q3

Six minutes. Name and matric number on the sheet; this is for feedback, not marks.

WRAP-UP · WHAT TO TAKE AWAY AND WHAT TO PREPARE

Next session: Galerkin, integration by parts, [k], assembly and the worked example

Today in four lines

- Energy balance + Fourier → (2.11): divergence of flux = source, written as $\{\partial\}[E]\{\partial\}^T T = -q_h$.
- Dirichlet = essential = imposed; Neumann = natural = appears from integration by parts; insulated = do nothing.
- Four-node element: $T = \{N\}\{\hat{T}\}^T$ with bilinear N_i; Kronecker delta, partition of unity, linear edges.
- $\{\partial\}^T\{N\}$ (2 × 4) gives the gradient and, with −[E], the flux — and the next session, [k].

Before Heat Transfer Part 2

- Read text §2.2.2 and §2.3 (worked example) — 12 pages.
- Redo Activities 4 and 5 for the point (0.5, 0.8) m.
- Classify the edges of the §2.3 plate and predict the hottest point.
- Bring MATLAB; Assignment 4 is released at the end of Heat Transfer Part 2.

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